SafeExit-RL: Adaptive Fire Evacuation Guidance under Partial Observability and Dynamic Congestion

Lokesh Motwani Jenish Thakkar

Department of Computer Science and Engineering Repository: `https://github.com/lokesh-2406/SafeExit-RL`

Abstract

Centralized, omniscient routing in building fire evacuation fails under realistic sensor degradation and crowd stampedes. We present **SafeExit-RL**, an adaptive evacuation framework formulated as a POMDP. We benchmark a Deep Q-Network (DQN) with a 14-dimensional continuous layout-invariant state and Relative Tabular Q-learning against Static Shortest Path (BFS) and Reactive Replanning (Dijkstra) across three regimes: (1) immediate egress, (2) 20-step pre-evacuation delay with stationary hazard exposure, and (3) zero-shot cross-layout transfer to an unseen multi-wing building. Under severe sensor noise (σ=0.3), DQN maintains 97.9% evacuation success and 2.59 cumulative hazard exposure, whereas Static Shortest Path degrades to 87.5% success with 3.74 exposure. Under a 20-step pre-evacuation delay, DQN achieves 100% success and 0.06 post-delay exposure, outperforming Reactive Replanning (95.0% success, 9.69 exposure). Finally, zero-shot transfer highlights the structural distinction between local perception-action RL and global topological search.

1 Introduction

Indoor fire evacuation requires fast, risk-averse routing under expanding flame fronts, toxic smoke, and moving evacuee crowds. Classical guidance relies on static exit signage or global shortest-path algorithms [1]. However, static paths can steer occupants into compromised halls, and central planners fail when communication links fail or fire blocks predicted routes [4].

While Reinforcement Learning (RL) enables dynamic route adaptation [3], prior work frequently presumes ideal sensing: full-grid hazard awareness without delay or noise. In real buildings, sensors have finite coverage (ρ=2), Gaussian noise (σ), and telemetry latency (τ). Moreover, occupant panic and herding generate dynamic choke points. **SafeExit-RL** addresses this via: (1) realistic sensing with noise, staleness, and radius limits; (2) crowd-coupled congestion; (3) rigorous pre-evacuation delay exposure accounting; and (4) coordinate-invariant representations for zero-shot transfer.

2 POMDP Problem Formulation

We formulate fire evacuation as a POMDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \Omega, \mathcal{O}, \gamma \rangle$. The environment state $s_t = (p_t, H_t, C_t, t) \in \mathcal{S}$ tracks occupant position $p_t \in \mathcal{G}$, true hazard field $H_t : \mathcal{G} \to [0,1]$, crowd positions $C_t : \mathcal{G} \to \mathbb{N}$, and simulation tick t. Within sensing radius ρ=2, the agent receives observation $o_t \in \Omega$:

$$o_t = \Big(p_t,\ \{\hat{h}_t(u)\}_{u\in\mathcal{N}_\rho},\ \{c_t(u)\}_{u\in\mathcal{N}_\rho},\ s_t,\ b_t,\ w_t\Big) \quad (1)$$

where $\mathcal{N}_\rho(p_t) = \{u \in \mathcal{G} \mid \|u - p_t\|_\infty \le \rho\}$. Observed hazard is corrupted by Gaussian noise: $\hat{h}_t(u) = \text{clip}(H_t(u) + \epsilon,\ 0,\ 1)$ with $\epsilon \sim \mathcal{N}(0, \sigma^2)$, refreshed every τ steps. Unobserved cells retain stale internal belief $B_t(u)$ with staleness $s_t(u) = t - t_{\text{last}}(u)$. Exit bearing $b_t = (b_{dr}, b_{dc}) \in [-1,1]^2$ is a unit vector pointing toward the nearest believed-safe exit. The action space $\mathcal{A} = \{\text{left}, \text{right}, \text{forward}, \text{backward}, \text{wait}\}$ comprises 5 discrete actions with deltas $(0,-1)$, $(0,1)$, $(-1,0)$, $(1,0)$, and $(0,0)$.

2.1 State Representations

1) DQN (14D Continuous): Feature vector $\phi(o_t) \in \mathbb{R}^{14}$ invariant to absolute spatial coordinates:

$$\phi(o_t) = \Big[w_t,\ b_t,\ \bar{h}_t,\ \hat{h}(p_t),\ \hat{h}_{\text{adj}},\ C_{\text{total}}/10,\ c(p_t)/5\Big] \quad (2)$$

where $w_t \in \{0,1\}^4$ is 4-directional local wall traversability (forward, backward, left, right); $b_t \in [-1,1]^2$ is exit bearing; $\bar{h}_t$ is mean believed hazard in $\mathcal{N}_\rho$; $\hat{h}(p_t)$ is believed hazard at p_t; $\hat{h}_{\text{adj}} \in [0,1]^4$ are cardinal neighbor hazards; $C_{\text{total}} = \sum_{u\in\mathcal{N}_\rho} c_t(u)$; and $c(p_t)$ is local crowd density at p_t.

2) Relative Tabular Q-learning: Discrete tuple $s_t^{\text{tab}} = (w_t, \text{bin}(\bar{h}_t), \text{bin}(C_{\text{total}}), \text{dir}(b_t))$ with 16 wall states (w_t), 4 hazard tiers ($[0, 0.05), [0.05, 0.2), [0.2, 0.5), [0.5, 1]$), 4 crowd tiers ($0, 1, 2, \ge 3$), and 5 bearing directions ($\text{dir}(b_t)$), totaling 1,280 states. Unseen states during evaluation default to $\text{dir}(b_t)$.

2.2 Reward Engineering

The scalar reward balances progress, hazard avoidance, congestion avoidance, and terminal exit arrival:

$$r_t = R_{\text{prog}} + R_{\text{haz}} + R_{\text{cong}} + R_{\text{term}} \quad (3)$$

where $R_{\text{prog}} = -1.0\cdot\mathbb{I}_{\text{wall}} - 0.3\cdot\mathbb{I}_{\text{revisit}} - 0.1\cdot\mathbb{I}_{\text{step}}$ penalizes wall collisions (-1.0), revisiting cells (-0.3), and elapsed time (-0.1); $R_{\text{haz}} = -B_t(p_t) \in [-1, 0]$ penalizes perceived hazard at the current cell using internal belief B_t; $R_{\text{cong}} = -0.2 \cdot c_t(p_t)$ penalizes occupancy in cells crowded by NPCs; and terminal exit reward is:

$$R_{\text{term}} = 20.0 \cdot \max(0,\ 1 - E_{\text{post}}) \cdot \mathbb{I}(p_t \in \mathcal{E}) \quad (4)$$

scaled strictly by post-delay ground-truth cumulative exposure $E_{\text{post}} = \sum_{k=1}^{T_{\text{post}}} H_k(p_k)$. Crucially, pre-evacuation exposure E_{pre} accumulated while stationary during delay does not attenuate R_{term}, preserving learning gradients for active egress routing.

3 Environment Dynamics and Baselines

The building is an $H \times W$ grid of walls (#), corridors ($\cdot$), fire origin (F), and exits (E). Fire spreads via an analyti-

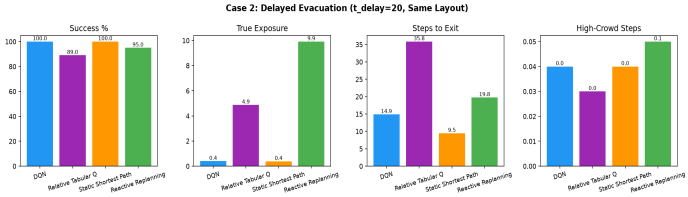


Figure 1: **Case 2 Benchmark ($t_{\text{delay}} = 20$ steps on `layouts/case1.txt`, 100 eps).** DQN achieves 100% success and 0.06 post-delay exposure, outperforming Reactive Replanning which oscillates in smoke (95% success, 9.69 exposure).

Table 1: Benchmark results across Cases 1, 2, and 3 (100 episodes per method, 5 seeds). Post-delay exposure (E_{post}) isolates active egress performance from stationary delay contamination.

Method	Succ.%	Post Exp	Tot Exp	Steps
Case 1: Immediate Evacuation ($t_{\text{delay}} = 0$, `layouts/case1.txt`)				
DQN (Ours)	**100.0**	**0.08**	**0.08**	17.4
Relative Tabular Q	100.0	0.33	0.33	23.9
Static Shortest Path	100.0	0.13	0.13	9.4
Reactive Replanning	100.0	0.12	0.12	10.1
Case 2: Delayed Evacuation ($t_{\text{delay}} = 20$, `layouts/case1.txt`)				
DQN (Ours)	**100.0**	**0.06**	**0.39**	14.9
Relative Tabular Q	89.0	4.66	4.88	35.8
Static Shortest Path	100.0	0.23	0.38	9.5
Reactive Replanning	95.0	9.69	9.91	19.8
Case 3: Zero-Shot Transfer ($t_{\text{delay}} = 0$, `layouts/case2.txt`)				
DQN (Ours)	25.0	48.42	48.42	151.9
Relative Tabular Q	24.0	57.19	57.19	153.4
Static Shortest Path	100.0	0.03	0.03	10.7
Reactive Replanning	99.0	1.83	1.83	12.9

cal radial sigmoid proxy: $r(t) = 20.0/(1 + e^{-0.05(t-60)})$, with ground-truth cell hazard $H_t(u) = \max(0,\, 1 - \|u - u_0\|_2 / r(t))$. $N = 10$ NPCs execute biased random walks toward nearest exits. If local hazard > 0.2, NPCs panic with probability p_{panic}, picking random directions; with probability p_{herd}, NPCs herd toward nearby peers, inducing dynamic blockages. Episodes end on exit arrival ($p_t \in \mathcal{E}$) or at $T_{\max} = 200$ active steps.

Primary Comparison Methods: (1) **DQN:** 2-layer MLP (hidden 64, ReLU), replay buffer 10^4, batch 64, $\gamma = 0.95$, target sync every 100 steps, Adam (lr $= 10^{-3}$), ϵ-decay $1.0 \to 0.05$ over 1,500 episodes, Huber loss. (2) **Relative Tabular Q:** Q-table over 1,280 states, $\alpha = 0.1$, $\gamma = 0.95$, ϵ-decay over 1,500 episodes. (3) **Static Shortest Path (Static SP):** Offline BFS at $t = 0$; executes blindly without replanning. (4) **Reactive Replanning:** Online Dijkstra replanning every $k = 5$ steps with edge costs $w(u, v) = 1.0 + 10.0 \cdot B_t(v) + 2.0 \cdot c_t(v)$.

4 Experimental Evaluation and Results

4.1 Case 1: Baseline Immediate Evacuation

Case 1 evaluates immediate egress ($t_{\text{delay}} = 0$) on `case1.txt` (9×30, 2 exits, fire origin $(4, 22)$) under sensor degradation ($\sigma = 0.1, \tau = 2$). As shown in Table 1, all four methods achieve 100% success. Graph baselines exploit global BFS to finish in 9.4–10.1 steps. However, DQN discovers an active risk-averse policy, navigating around hazardous corridor fringes to achieve lower exposure (0.08) than Static SP (0.13).

4.2 Case 2: Delayed Evacuation with Corridor Engulfment

Case 2 imposes a 20-step pre-evacuation delay, during which fire expands from $r = 0.95$ m to 2.38 m, engulfing the hallway. DQN demonstrates superior resilience (Table 1, Fig. 1): 100% success and 0.06 post-delay exposure (total 0.39). Reactive Replanning wavers due to belief staleness (95.0% success, 9.69 post-delay exposure), while Relative Tabular Q drops to 89.0% success.

4.3 Case 3: Zero-Shot Cross-Layout Transfer

Case 3 evaluates policies trained on Case 1 zero-shot on an unseen layout (`case2.txt`, 13×36, 4 exits, multi-wing corridors) with zero retraining.

DQN achieves 25.0% and Relative Tabular Q achieves 24.0% success (Table 1). Both learned methods transfer local navigation skills (following exit bearings, avoiding immediate walls), but fail when non-convex dead ends require backtracking beyond sensing radius ρ=2. In contrast, Static SP (100%) and Reactive Replanning (99%) succeed by computing global paths over the known topology, highlighting the boundary between local perception-action RL and global graph search.

4.4 Sensor Degradation Sensitivity Sweep

Sweeping noise ($\sigma \in [0.0, 0.3]$) and latency ($\tau \in [0, 10]$) on Case 1 demonstrates RL robustness. At $\sigma = 0.3$, Static SP degrades to 87.5% success with 3.74 exposure (Reactive: 4.21 exposure), whereas DQN sustains 97.9% success with 2.59 exposure. Under crowd panic and herding ($p_{\text{panic}} = p_{\text{herd}} = 0.3$), DQN maintains 0.09 exposure (vs. 0.70 for tabular Q), confirming the efficacy of its continuous congestion features.

5 Limitations and Physical Disclaimer

Hazard Model: The sigmoidal radial model is an analytical tenability proxy ($O(1)$ query complexity), not a numerical CFD/FDS fire simulation or validated Fractional Effective Dose (FED) model.

Delay Parameterization: The 20-step delay is a chosen simulation parameter designed to induce corridor bottlenecks. Time conversion (1 step $\approx$ 1 s) is an unvalidated nominal assumption based on 1.0 m cells and 1.0 m/s walking speed.

6 Conclusion

SafeExit-RL demonstrates that deep reinforcement learning with continuous relative state representations achieves robust, risk-averse fire evacuation guidance under degraded sensing and dynamic corridor bottlenecks where classical heuristics fail. Future work will combine local RL perception with global topological memory.

References

[1] H. Fang, P. Lin, and S. Gao, "Development of a tenability-based path planning model for building fire evacuations using reinforcement learning," *Journal of Building Engineering*, vol. 118, p. 115132, 2026.

[2] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.

[3] C. Chu, J. Chen, and Q. Xu, "Reinforcement learning based pedestrian evacuation with social force model," in *Proc. IEEE ICRA*, 2019, pp. 7532–7538.

[4] D. Silver and J. Veness, "Monte-Carlo planning in large POMDPs," in *NeurIPS*, vol. 23, 2010, pp. 2164–2172.

[5] Z. Shi, E. Zlatanova, and A. Kievits, "Multi-agent reinforcement learning for building evacuation under dynamic fire hazards," *Automation in Construction*, vol. 132, p. 103930, 2021.

[6] D. Helbing and P. Molnar, "Social force model for pedestrian dynamics," *Physical Review E*, vol. 51, no. 5, pp. 4282–4286, 1995.